CSE 400: Fundamentals of Probability in Computing
Lecture 20 Scribe

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1 Random Vector Representation

1.1 Definition

For random variables $X_1, X_2, \ldots, X_N$, we represent them as a vector:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- X is called a random vector.
- It provides a compact representation of multiple random variables.

2 Mean Vector

2.1 Definition

The mean vector of X is:

$$\mu_X = E[X]$$

- Represents the average value of the random vector.
- Each element corresponds to the mean of one random variable.

2.2 Structure

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

3 Correlation Matrix

3.1 Definition

$$R_{XX} = E[XX^T]$$

$$R_{XX}(i,j) = E[X_i X_j]$$

3.2 Matrix Representation

$$R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \cdots & E[X_1X_N] \\ E[X_2X_1] & E[X_2X_2] & \cdots & E[X_2X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_NX_1] & E[X_NX_2] & \cdots & E[X_NX_N] \end{bmatrix}$$

- R_{XX} is a symmetric matrix.

4 Covariance Matrix

4.1 Definition

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

4.2 Matrix Form

$$C_{XX} = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- $\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$

5 Linear Transformation of Random Vectors

5.1 Model

$$Y = AX + b$$

- A: Transformation matrix
- b: Constant vector
- X: Input random vector

5.2 Expanded Form

$$\begin{aligned}
Y_1 &= a_{11}X_1 + a_{12}X_2 + \cdots + a_{1N}X_N + b_1 \\
Y_2 &= a_{21}X_1 + a_{22}X_2 + \cdots + a_{2N}X_N + b_2 \\
&\vdots \\
Y_N &= a_{N1}X_1 + a_{N2}X_2 + \cdots + a_{NN}X_N + b_N
\end{aligned}$$

6 Mean of Transformed Vector

6.1 Definition

$$\mu_Y = E[Y]$$

6.2 Derivation

$$\begin{aligned}
\mu_Y &= E[AX + b] \\
&= E[AX] + E[b]
\end{aligned}$$

6.3 Result

$$\mu_Y = A\mu_X + b$$

7 Correlation Matrix Transformation

7.1 Definition

$$R_{YY} = E[YY^T] = E[(AX + b)(AX + b)^T]$$

7.2 Expansion

$$R_{YY} = E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$$

7.3 Applying Expectation

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X]^TA^T + bb^T$$

7.4 Final Result

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

8 Covariance Matrix Transformation

8.1 Key Observation

$$Y - \mu_Y = (AX + b) - (A\mu_X + b) = A(X - \mu_X)$$

8.2 Derivation

$$\begin{aligned} C_{YY} &= E[(Y - \mu_Y)(Y - \mu_Y)^T] \\ &= E[A(X - \mu_X)(X - \mu_X)^T A^T] \end{aligned}$$

8.3 Result

$$C_{YY} = AC_{XX}A^T$$

- The shift b does not affect covariance.

9 Summary

If $Y = AX + b$, then:

- Mean:
$$\mu_Y = A\mu_X + b$$
- Covariance:
$$C_{YY} = AC_{XX}A^T$$
- Correlation:
$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

10 Example: Numerical Linear Transformation

Given:

$$\mu_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \quad R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

10.1 (a) Mean Vector

$$\mu_Y = A\mu_X$$

$$= \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

10.2 (b) Correlation Matrix

Step 1:

$$AR_{XX} = \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix}$$

Step 2:

$$R_{YY} = (AR_{XX})A^T = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$$

10.3 (c) Covariance Matrix

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

$$\mu_Y \mu_Y^T = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

$$C_{YY} = \begin{bmatrix} 17 & -6 & -9 \\ -6 & 14 & -8 \\ -9 & -8 & 19 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

11 Gaussian Distributions

11.1 Univariate Gaussian

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

11.2 Joint Gaussian (2D)

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} \exp\left(-\frac{1}{2(1-\rho_{XY}^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) +\right.\right.$$

12 Multivariate Gaussian Distribution

12.1 Definition

$$X = [X_1, X_2, \ldots, X_N]^T$$

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

- μ_X: Mean vector
- C_{XX}: Covariance matrix

13 2D Gaussian Derivation

13.1 Determinant

$$\det(C_{XX}) = \sigma_1^2\sigma_2^2 - (\rho\sigma_1\sigma_2)^2 = \sigma_1^2\sigma_2^2(1-\rho^2)$$

13.2 Inverse

$$C_{XX}^{-1} = \frac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)}\begin{bmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{bmatrix}$$

13.3 Quadratic Form

$$Q(x) = (x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)$$

$$Q(x) = \frac{1}{1-\rho^2}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]$$

14 Uncorrelated Gaussian Variables

14.1 Result

If $\text{Cov}(X_i, X_j) = 0$ for $i \neq j$, then:

$$f_X(x) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$$

- Covariance matrix is diagonal.
- Joint PDF factorizes into product of individual PDFs.

15 Homework

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$